USER INTERFACE DESIGN

EXPERIEMENT-2

Design a UI where users recall visual elements (e.g., icons or text chunks).
Evaluate the effect of chunking on user memory.

Introduction:

Chunking is a cognitive strategy that involves breaking information into smaller, meaningful units. This experiment uses a game-based UI to analyze how chunking improves short-term memory recall by presenting visual elements within a limited time.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis

The instruction page effectively applies chunking by presenting information in a structured and organized manner.

1. **Clear Sequential Steps**
   Instructions are divided into numbered steps, making them easy to understand and follow.

2. **Logical Grouping**

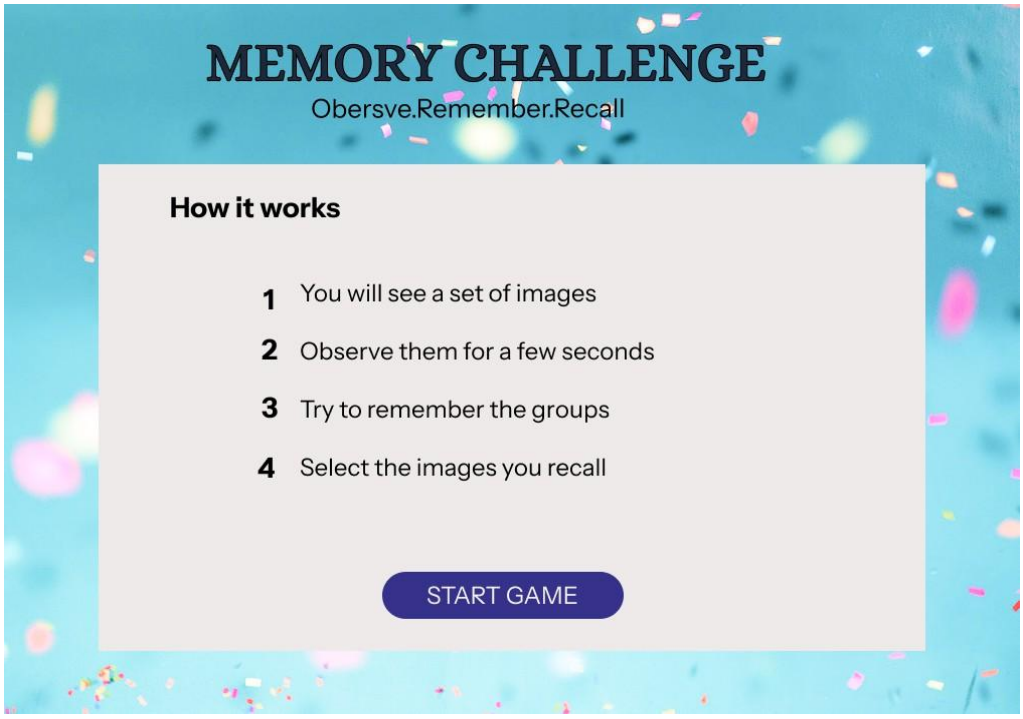
   - Observation Phase – explains what users will see.
   - Memorization Phase – encourages focusing on remembering items.
   - Recall Phase – explains how users should recall items.

- Performance Metric – accuracy is emphasized over speed.

3. **Visual Hierarchy**
   A bold title, proper spacing, and a highlighted START button guide the user clearly.

4. **Time Awareness**
   Users are informed that they have only 5 seconds, preparing them to focus.

5. **Simplicity**
   Short sentences and minimal text reduce cognitive load.



FRAME 2: CHUNKING PHASE

Analysis of Chunking Phase Screen

Purpose:

This is the memory encoding stage where users observe icons and mentally group them.

UI Elements:

- Countdown Timer (5 seconds) to create urgency.
- Progress bar to visualize remaining time.
- Grid of icons arranged to support grouping by category, color, or similarity.

Working:

Users quickly scan and form mental chunks before time ends.

Cognitive Benefit:

Chunking reduces memory load and improves recall efficiency.

FRAME 3: RECALL PHASE

Analysis of the Recall Phase

1. Purpose

- Tests memory by asking users to recall previously seen icons.
- Measures the effectiveness of chunking.

2. UI Elements

- **Title & Instructions:** Clearly explain the task.
- **Multiple Choice Icon Grid:**
  - Contains both correct icons and distractors.
  - Users must identify previously seen items.
- **Radio Buttons:**
  - Easy and clear selection method.
- **Submit Button:**
  - Confirms user choices.
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3. Working Principle

1. Users recall icons from memory.
2. Select remembered icons.
3. Distractors test memory accuracy.
4. Submit leads to result page.

4. UX & Cognitive Benefits

- Evaluates true recall ability.
- Reduces guessing through distractors.
- Gamified design increases engagement.

FRAME 4: RESULT PAGE

Analysis of the Result Page

1. Purpose

- Displays recall accuracy.
- Provides feedback and next actions.

2. UI Elements

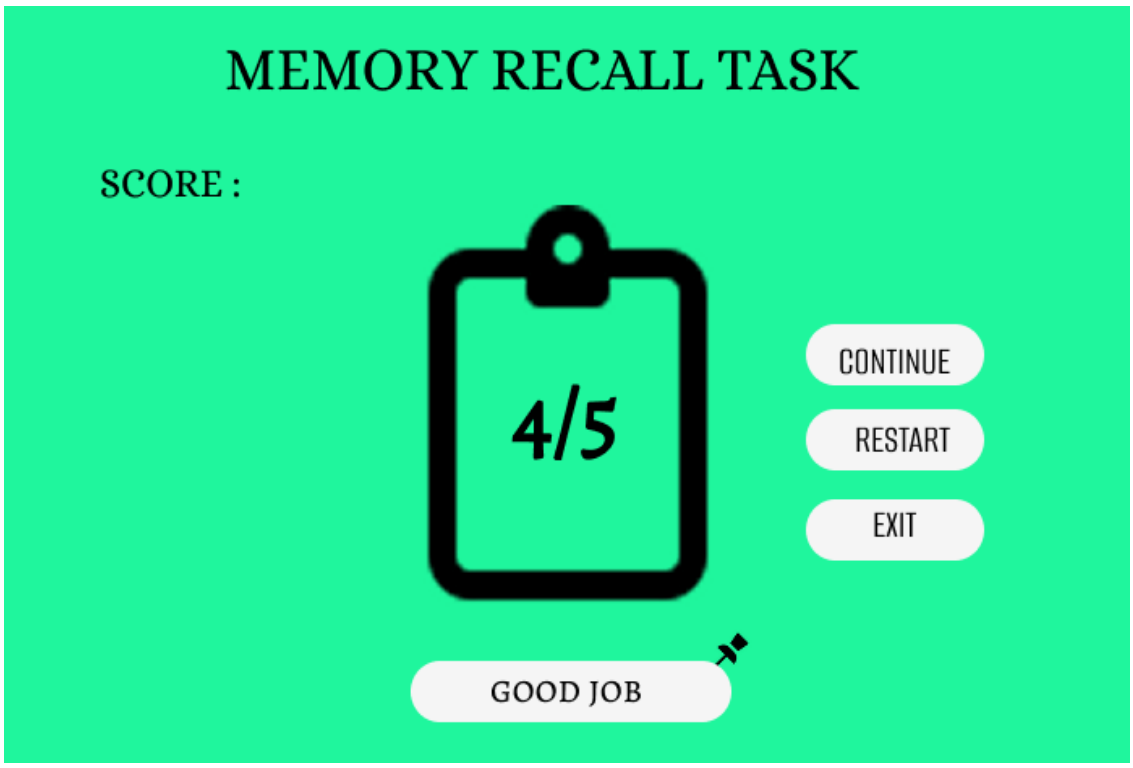
- **Score Display (e.g., 4/5):**
  - Shows performance clearly.
- **Action Buttons:**
  - **Play again:** Move to next level.
  - **Home:** End the activity.

3. Working Principle

1. System evaluates selected icons.
2. Calculates score.
3. User chooses next action.

4. Cognitive & UX Benefits

- Immediate feedback improves learning.
- User control enhances experience.
- Gamification reduces stress and increases motivation.



PROTOTYPE LINK:

https://www.figma.com/proto/LDMbQ20Mp3jROcxuybs0Sc/Untitled?node-id=0-1&t=4fuXeInIlRtK6v9M-1

https://www.figma.com/proto/r1nao5fXI2kWy2KZlcTWcK/Untitled?t=4fuXeInIlRtK6v9M-1

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